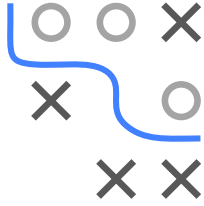
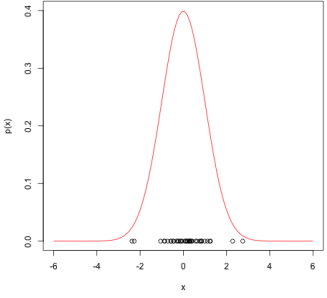
Introduction to Deep Learning



Generative Adversarial Networks
Introduction to Generative Models

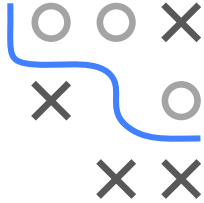


Learning goals

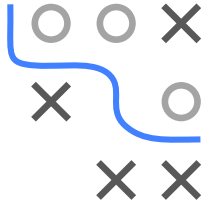
- learning a generative model
- examples of generative models

WHICH FACE IS FAKE?

Fake faces generated by ThisPersonDoesNotExist.com. | Image: The Verge



DEEP UNSUPERVISED LEARNING

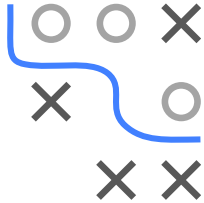


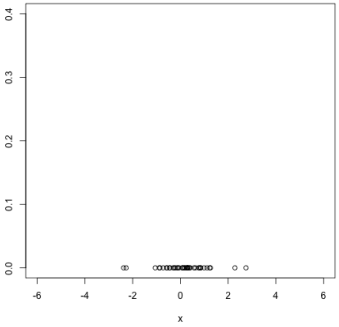
There are two main goals of **deep unsupervised learning**:

- **Representation Learning**
  - Examples are: manifold learning, feature learning, etc.
  - Can be done by an autoencoder
  - Examples of applications:
    - dimensionality reduction / data compression
    - transfer learning / semi-supervised learning
- **Generative Models**
  - Given a training set $\mathcal{D} = (\boldsymbol{x}^{(1)}, \ldots, \boldsymbol{x}^{(n)})$ where each $\boldsymbol{x}^{(i)} \sim \mathbb{P}_x$, the goal is to estimate $\mathbb{P}_x$.
  - **Goal:** Take as input training samples from some distribution and learn a model that represents that distribution!
  - Examples of applications:
    - generating music, videos, volumetric models for 3D printing, synthetic data for learning algorithms, outlier identification, images denoising, inpainting, etc.

DENSITY FITTING / LEARNING A GENERATIVE MODEL

Given $\mathcal{D} = \left(\boldsymbol{x}^{(1)}, \boldsymbol{x}^{(2)}, \ldots, \boldsymbol{x}^{(n)}\right) \sim \mathbb{P}_x$ learn a model of $\mathbb{P}_x$ (for example, fitting a Gaussian distribution via Maximum Likelihood Estimation).





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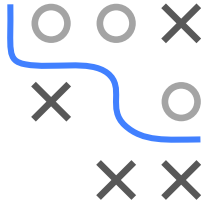
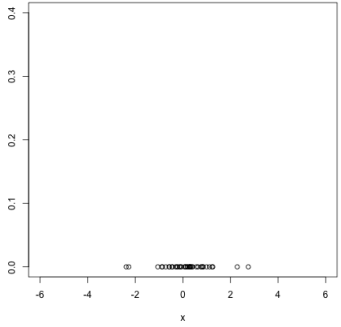
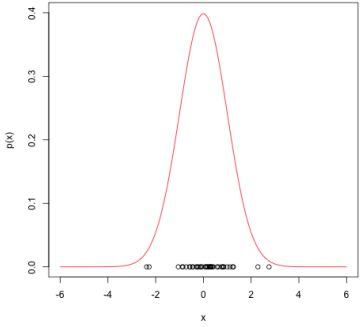


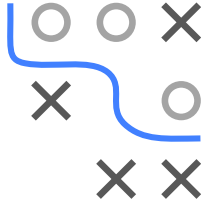




WHY GENERATIVE MODELS?

Generative model are capable of uncovering underlying latent variables in a dataset and can be used for

- sampling / data generation
- outlier detection
- missing feature extraction
- image denoising / reconstruction
- representation learning
- planning in reinforcement learning
- ...

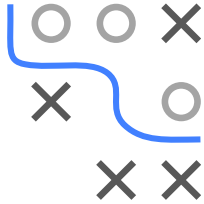


APPLICATION EXAMPLE: IMAGE GENERATION

Source: Karras et al. (2018)

Figure: Synthetic faces generated by a Generative Adversarial Network (more on this later).

APPLICATION EXAMPLE: NEURAL STYLE TRANSFER



A photograph is “redrawn” in the style of another image! (Gatys et al., 2015)

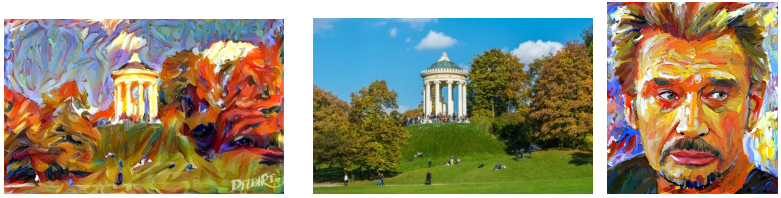


Figure: Examples generated on `https://deepart.io/`. The image on the left has been generated by translating the original image (middle) to the style of the image on the right.

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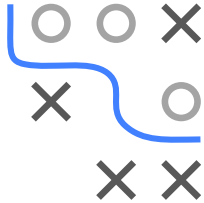

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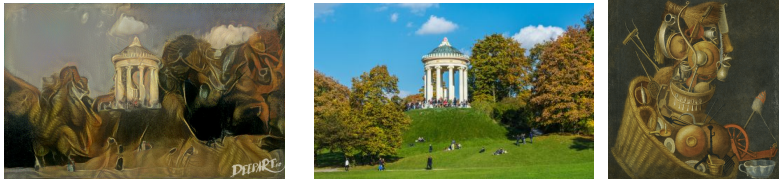

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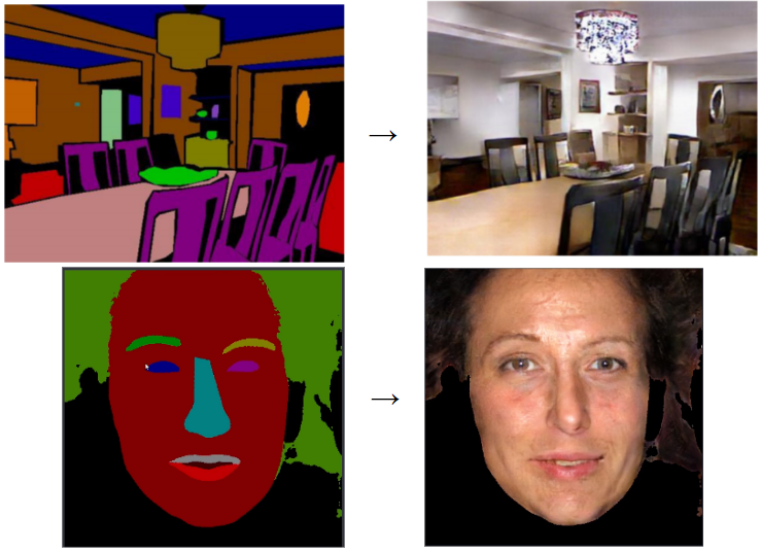
APPLICATION EXAMPLE: IMAGE INPAINTING

Source: Demir et al (2018)

Figure: A generative model fills in the missing portion of the image based on the surrounding context.

APPLICATION EXAMPLE: SEMANTIC LABELS –> IMAGES

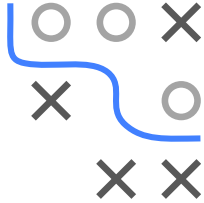


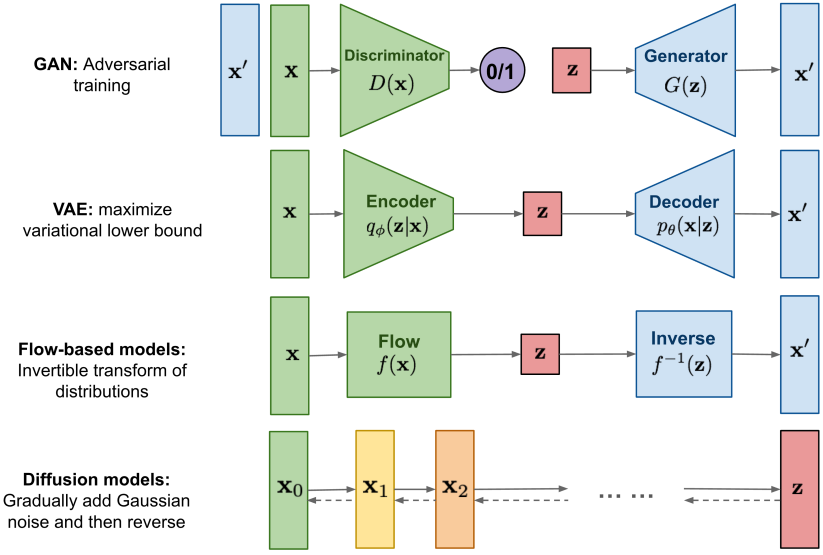
Source: Wang et al (2017)

APPLICATION EXAMPLE: GENERATING IMAGES FROM TEXT

Source: Zhang et al (2017)

OVERVIEW GENERATIVE MODELS





Source: lilianweng.github.io

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https://arxiv.org/abs/1710.10196

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Neural Algorithm of Artistic Style
https://arxiv.org/abs/1508.06576